What makes motion hard to stop?

antocciphysics | TEACHER KEY | CPA

Learning target: Compare momentum changes in a controlled stopping situation.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 14; 3 course days later.*
Fastest at 3 AU. Sweeping equal areas in equal times requires greater speed near the Sun.

02. Explain a worked example (worked)
Velocity arrows equal; momentum arrow for 4 kg cart twice as long.

03. Look and explain (model)
B: 6 kg·m/s versus A: 2 kg·m/s; velocity is held fixed.

04. Build the solution (practice)
A 2 N; B 6 N. Both stopping forces point left.

05. Use the idea (reason)
No. Stopping time, energy, deformation, contact area and material response matter.

06. Choose a method (mixed) *Review: Unit 2, Day 5; 12 course days later.*
4 N, attractive toward the other mass.

07. Exit check (exit)
6 kg·m/s and 6 N.

08. STRETCH: Apply it another way (stretch)
Example 2 kg at 6 m/s gives 12 kg·m/s right.

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01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 14; 3 course days later.*
$T_B/T_A=3^{(3/2)}\approx5.196$. The normalization depends on the central mass and chosen units.

02. Explain a worked example (worked)
Velocity arrows equal; momentum arrow for 4 kg cart twice as long.

03. Represent and interpret (model)
B: 6 kg·m/s versus A: 2 kg·m/s; velocity is held fixed.

04. Solve and justify (practice)
Δp=-2,-6,-4 kg·m/s; forces=-4,-12,-8 N.

05. Analyze the reasoning (reason)
No. Stopping time, energy, deformation, contact area and material response matter.

06. Choose a method (mixed) *Review: Unit 2, Day 5; 12 course days later.*
$F\approx2.964\times10^{-5}$ N, attractive. The incorrect numerical answer is 6 times too large when SI values are used.

07. Exit check (exit)
J=-12 N·s; average F=-6 N.

08. STRETCH: Apply it another way (stretch)
Each momentum +4 kg·m/s; change -4; average force -20 N. Equal momentum change over equal time gives equal average force.